Xinbao Qiao

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Education

The Chinese University of Hong Kong Ph.D., Information Engineering. Advisor: Angela Yingjun Zhang. *Aug 2026∼present*

Research Areas My research primarily studies lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted. My recent work aims to improve the reliability, interpretability, and controllability of AI models in heterogeneous, computation-constrained, and communication-constrained environments.

Zhejiang University M.Eng. in Artificial Intelligence. Advised by Meng Zhang. *Zhejiang. Sep 2022 ∼ Dec 2025*

Shandong University B.Eng. in Communication Engineering. *Shandong. Sep 2018 ∼ Jul 2022*

Experience

Doctoral Research @ The Chinese University of Hong Kong *Hong Kong. Aug 2026 ∼ present*

Investigates data lifecycle management in distributed AI across generation, use, deletion, and federated learning.

- Proposed optimal-transport federated learning with barycentric multi-prototype classification. Paper #6

Master's Research on Data-Centric ML Systems @ Zhejiang University *Zhejiang. Mar 2023 ∼ Dec 2025*

Developed data-deletion methods for continuous influence control, certified online removal, and dynamic tree ensembles.

- Extended machine unlearning beyond binary erasure through continuous example weights, enabling targeted fairness and robustness interventions. Paper #3
- Built Hessian-free online certified unlearning with recollected trajectory statistics, replacing explicit Hessian inversion for streaming deletion requests. Paper #4
- Built exact dynamic random-forest unlearning through affected-tree updates, avoiding full retraining. Paper #5

Research Internship on Trustworthy LLM Systems @ NUSRI CQ *Chongqing. Jun 2025 ∼ Dec 2025*

Studied LLM reliability in synthetic-data feedback loops and unsupported relational inference.

- Showed how low-resource verification amplifies sample-selection bias in recursive synthetic-data training, persistently pruning tail data and precipitating model collapse. Paper #1
- Compared LLM and human illusory pattern perception across three psychological tasks and analyzed associated SAE features. Paper #2

Open-Source Contributions and Academic Service

- **Research code releases**, public code for certified unlearning, soft-weighted unlearning, and model-collapse work.
- **Xinbaopedia**, public academic homepage and wiki-style research archive with papers, figures, CV, and project notes.
- **Academic service, 2026**, reviewer for ICML, NeurIPS, and AAAI; **NeurIPS 2026 Top Reviewer**.
- **Academic service, 2025**, reviewer for NeurIPS, ICLR, AAAI, and IEEE TNNLS.

Selected Publications

Asterisks (*) denote co-first authorship; daggers (†) denote corresponding authors.

1. **When Sample Selection Bias Precipitates Model Collapse** *ICML 2026.*
   Xinbao Qiao†, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang†, Yan Pang†
   Shows how low-resource verification turns selection bias into persistent tail pruning and model collapse.
2. **Illusory Pattern Perception Drives Spurious Inference in Large Language Models** *NeurIPS 2026.*
   Peihua Mai*, Zhuoyan Shao*, **Xinbao Qiao***, Meng Zhang, Xinyue Zhou†, Yan Pang†
   Compares LLM and human illusory pattern perception with SAE analysis of associated features.
3. **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness** *AAAI 2026.*
   Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang†
   Generalizes binary deletion into continuous influence control for targeted fairness and robustness interventions.
4. **Hessian-Free Online Certified Unlearning** *ICLR 2025.*
   Xinbao Qiao, Meng Zhang†, Ming Tang, Ermin Wei
   Replaces explicit Hessian inversion with recollected trajectory statistics for scalable certified deletion.
5. **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest** *ICLR 2025.*
   Shurong Wang, Zhuoyang Shen, **Xinbao Qiao**, Tongning Zhang, Meng Zhang†
   Updates affected tree statistics to deliver exact random-forest unlearning in dynamic environments.
6. **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification** *Under review.*
   Xinbao Qiao, Wenjing Yan†, Ying-Jun Angela Zhang
   Frames federated learning as optimal transport with barycentric multi-prototype classification under communication constraints.
7. **Decentralized Free-Support Wasserstein Barycenter** *Under review.*
   Xinbao Qiao, Bokai Hou, Peihua Mai, Wenqian Li, Wenjing Yan, Ying-Jun Angela Zhang
   Learns movable barycenter supports through neighbor gossip, separating geometric quality from network agreement.